AJKD Atlas of Renal Pathology: Diffuse Mesangial Sclerosis

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Clinical and Pathologic Features

Diffuse mesangial sclerosis presents with nephrotic syndrome at birth or within the first year of life. It may occur as part of Denys-Drash syndrome or as an isolated lesion. Patients with Denys-Drash syndrome typically are 46XY with ambiguous or female external genitalia, and streak (undeveloped) gonads or testes. Progressive kidney disease develops, usually by age 4 years. Patients have increased risk for Wilms tumor.

Light microscopy: Glomeruli are small and condensed in appearance, with early lesions showing increased loose mesangial collagen that progress to sclerosis with dense collagen without hypercellularity. Podocytes do not show hyperplasia but may be immature and cobblestone-like.

Immunofluorescence microscopy: No specific staining.

Electron microscopy: Extensive foot process effacement without deposits, but increased collagen within mesangial areas.

Etiology/Pathogenesis

Patients with either isolated or syndromal forms of diffuse mesangial sclerosis have mutations in the *WT1* gene, which encodes a transcription factor.

Differential Diagnosis

Tubules may occasionally be dilated when extensive sclerosis develops, but this is not an early or dominant feature, as seen in congenital nephrotic syndrome of Finnish type. Congenital nephrotic syndrome of Finnish type is also distinguished by lack of dense collagenous material within the mesangium.

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Key Diagnostic Features

- Small, condensed glomeruli with increased dense mesangial collagen
- Extensive foot process effacement

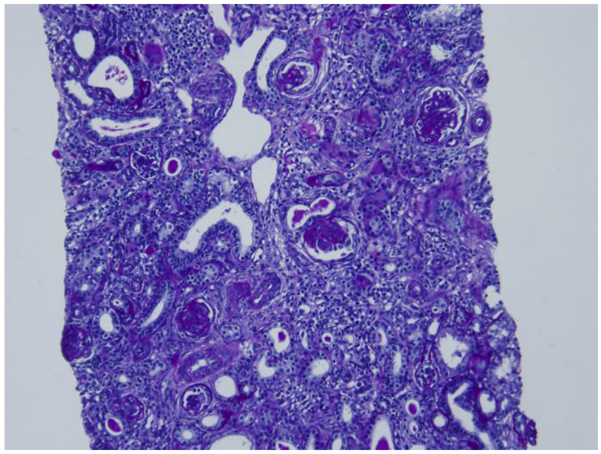

Figure 1. Diffuse mesangial sclerosis with glomerular sclerosis and interstitial fibrosis (periodic acid–Schiff stain).

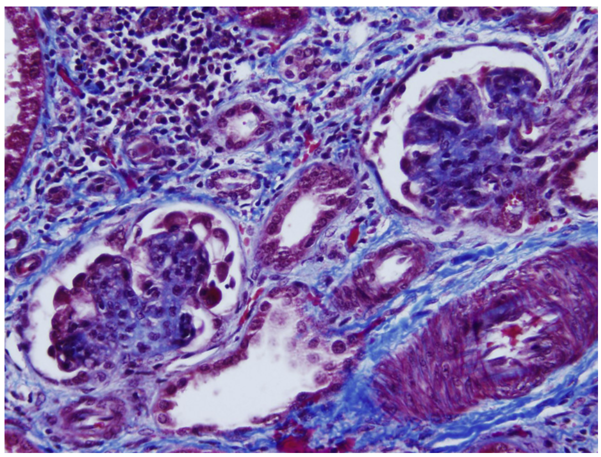

Figure 2. Diffuse mesangial sclerosis with typical-appearing dense collagenous material within the mesangium (Masson trichrome stain).

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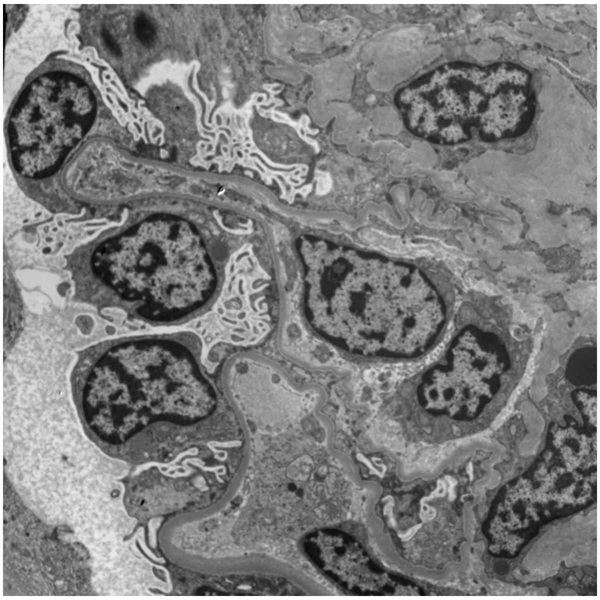

Figure 3. Diffuse mesangial sclerosis with extensive foot process effacement and no immune complex deposits (electron microscopy).